DeepScientist

AI REVIEW REPORT

AIG_CUR_2026_FARS_citation_consistent_voting

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Document No.	a2038411-330f-449a-abd3-bb05579d3463	Work ID	DS-RV-A20384113
Status	Completed	Agent Model	gpt-5.2
Token Usage	Input 9508755 | Output 122359 | Total 9631114	LLM Requests	63
Generated At	2026-06-09 16:06:14 UTC	Producer	DeepReviewer 2.0

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Final Agent Report

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Summary

Paper type: **training-free inference-time method + empirical evaluation**.

This paper proposes **Citation-Consistent Voting (CCV)** for permutation-ensembled RAG: run the same retrieved top-`N` document set under `K` random orderings, prompt the generator to emit a structured citation, map the cited index back to a **permutation-invariant document ID**, and select the answer whose citations are most consistent across runs (Page 3 - Method: 3.1–3.3; Page 4 - Eq. (1)–(2)). On NaturalQuestions with Qwen3-8B + Contriever top-5 retrieval, CCV reports a **+0.19 SubEM** gain over answer-string majority voting at `K=20` (Page 4 - Experiments: 4.2).

The core idea—using **evidence identity** (docID agreement) rather than **answer-string frequency** as the voting signal—is intuitive and potentially useful as a drop-in aggregator. However, the current manuscript is not yet fully defensible because key parts of the method are underspecified (edge cases/ties; answer canonicalization; docID/locID inconsistency), the "evidence-validity" filter is overly extractive and can bias results, and the empirical evidence is too narrow to support broad robustness claims (single dataset/model/retriever; greedy decoding only; small deltas without robust uncertainty reporting). I view this as a **promising but incremental** contribution that requires **major revisions** focused on correctness/reproducibility and a stronger, more decision-relevant evaluation package.

Strengths

- **Timely and practically motivated failure mode.** The paper targets *retrieval-order / permutation sensitivity* in RAG (Page 1 - Abstract; Page 1 - Introduction), which is a real reliability issue in production pipelines where retrieval order can change due to upstream nondeterminism.
- **Drop-in, training-free aggregation mechanism.** CCV's core design—map per-run citations to a permutation-invariant document ID and vote on evidence identity rather than answer-string frequency—is simple and easy to implement on top of existing "run `K` permutations then aggregate" pipelines (Page 3 - Method: 3.2–3.3; Page 4 - Eq. (1)–(2)).
- **Includes signal diagnostics and a useful ablation hook.** The paper does more than report one number: it analyzes citation-agreement vs correctness (Page 6 - Experiments: 4.5) and studies the trade-off between strict quote verification vs higher coverage (Page 4 - Practical Considerations; Page 5 - Experiments: 4.4).
- **The manuscript is compact and easy to follow.** The 3-stage pipeline and scoring rule are communicated succinctly, which makes the contribution accessible to readers who want a minimal "inference-only" robustness trick (Page 2 - Related Work; Page 3 - Method).

Weaknesses

1. **Method definition has reproducibility gaps and notation inconsistencies.** The manuscript uses inconsistent identifiers (docID vs locID) and leaves key edge cases unspecified (e.g., what if *no* runs pass evidence filtering; tie-handling for $\texttt{doc}^*\texttt{(a)}$ and `score(a)`). These silent choices can change outcomes when gains are small (Page 3 - Method: 3.2–3.3; Page 4 - Eq. (1)–(2)).

2. “Evidence-validity” filtering is overly extractive and can bias results. Requiring the *answer string to appear in the quote* is not a general grounding check; it rejects many correct but abstractive answers and can be gamed by copying. The paper’s own ablation suggests relaxing the quote constraint matters a lot, which further highlights sensitivity to this design (Page 3 - Method: 3.2; Page 4 - Practical Considerations; Page 5 - Experiments: 4.4).
3. **Answer-string grouping without canonicalization undermines the voting premise.** CCV groups by “unique answer” and tie-breaks by raw frequency, but QA answers have many surface-form aliases; without normalization/equivalence classes, both majority vote and CCV are artificially weakened and the comparison can be unstable (Page 3 - Method: 3.3; Page 4 - Eq. (1)–(2)).
4. **Empirical evidence is narrow and the claimed robustness/general compatibility is overbroad.** Experiments are limited to NaturalQuestions with one retriever (Contriever) and one generator (Qwen3-8B) under greedy decoding (T=0). This is insufficient to support “compatible with any RAG system…” or to argue robustness beyond this exact setting (Page 1 - Abstract; Page 4 - Experiments: 4.1).
5. **Small headline gains are not backed by adequate uncertainty and metric-sensitivity audits.** The main improvement is +0.19 SubEM at `K=20`. SubEM is substring-based and sensitive to formatting/normalization, yet the paper does not report normalization rules, paired significance tests, or repeated permutation-seed variance. There is also at least one K-dependent numerical ambiguity in the quote-constraint section that needs reconciliation (Page 4 - Experiments: 4.2; Page 5 - Experiments: 4.3–4.4).
6. **Related-work positioning is underdeveloped for closest neighbors.** The paper does not clearly contrast CCV with contemporaneous order-invariance or permutation-robust methods (e.g., Stable-RAG [1], KV-Fusion [2]) and attribution-faithfulness work (e.g., MIRAGE [3]), nor does it acknowledge self-citation failure modes as a core risk to a citation-based selector (Page 2 - Related Work).

Key Issues

Ranked Error Board (Top-10)
(Severity | Research-Value Impact | Validity Risk | Fixability | Confidence)

1. **Major | High | High | High | High** — *Evidence-validity filter is extractive and biases the selector.* Requires answer substring-in-quote; rejects abstractive correct answers and can be gamed (Page 3 - Method: 3.2; Page 5 - Experiments: 4.4).
2. **Major | High | High | High | High** — *Algorithm is not fully specified (edge cases/ties) + docID/locID inconsistency.* Missing fallback when no valid runs; inconsistent identifier naming; unclear argmax domain (Page 3 - Method: 3.3; Page 4 - Eq. (1)–(2)).
3. **Major | Medium | Medium | High | High** — *Answer-string grouping without canonicalization undermines both CCV and baselines.* Needs normalization/equivalence classes for fair aggregation (Page 3 - Method: 3.3).
4. **Major | High | Medium | Medium | High** — *Empirical evidence too narrow for broad robustness/compatibility claims.* Single dataset/model/retriever; greedy decoding only; no cross-setting validation (Page 1 - Abstract; Page 4 - Experiments: 4.1).
5. **Major | High | Medium | Medium | Medium-High** — *Small headline delta (+0.19 SubEM) lacks robust uncertainty + metric-sensitivity auditing.* Need answer normalization disclosure, paired significance/bootstraps, and repeated permutation-seed variance; also reconcile K-dependent numeric ambiguities (Page 4 - Experiments: 4.2; Page 5 - Experiments: 4.3–4.4).

6. **Major | Medium | Medium | Medium | Medium** — Causal interpretation of "monotonic with K" not established. Needs signal-level evidence (AUC/effect size) and multi-seed K curves (Page 5 - Experiments: 4.3).
7. **Major | Medium | Medium | Medium | Medium** — *Diagnostic analysis uses p-value without effect size/calibration.* Significance at n=3610 is not enough; need effect size + accuracy-vs-agreement calibration (Page 6 - Experiments: 4.5).
8. **Major | Medium | Low-Med | High | High** — *Related-work positioning underdeveloped vs closest neighbors.* Must contrast clearly with permutation-robust and attribution-faithfulness methods; acknowledge self-citation failure modes (Page 2 - Related Work).
9. **Minor | Medium | Low | High | Medium** — *Threat model definition is blurry (long-context position bias vs short-context permutation sensitivity).* Clarify robustness target metric (flip rate/variance) early (Page 1 - Introduction; Page 2 - Related Work: 2.1).
10. **Minor | Medium | Low | Medium | Medium** — *Compute/latency trade-off not quantified.* CCV requires K forward passes; report cost and consider "unstable-query-only" triggering (Page 4 - Method: 3.4; Page 6 - Conclusion).

Page Coverage Audit

Page	Annotation Count	Coverage Status	Skip Reason
Page 1	5	covered	—
Page 2	4	covered	—
Page 3	3	covered	—
Page 4	4	covered	—
Page 5	2	covered	—
Page 6	2	covered	—

ASCII Diagram — Paper Structure & Evidence Map

```
[Problem]
  Permuting a fixed retrieved doc set changes RAG answers
      |
      v
[Proposed Signal]
  docID agreement across permutations via citations
      |
      v
[Algorithm]
  Filter -> map doc_idx to docID -> score by modal-docID count -> select
      |
      +--> (Gap A) evidence-validity filter is extractive / underspecified
      |
      +--> (Gap B) missing answer normalization + tie/edge-case rules
      v
[Evidence]
  NQ (3610) + Qwen3-8B + Contriever top-5, T=0
  +0.19 SubEM vs majority vote at K=20
  agreement mean diff 3.04 vs 2.85 (p-value)
      |
      +--> (Gap C) uncertainty/statistics + metric-sensitivity not shown
      +--> (Gap D) external validity (datasets/models/decoding) not shown
```

Actionable Suggestions

Must (publication-critical)

1. Replace the extractive “answer-in-quote” evidence filter with a normalization-aware grounding check.
   · Where: Page 3 - Method: 3.2.
   · What to implement (training-free): canonicalize answer/quote strings (case-fold, punctuation stripping, whitespace normalize), plus domain rules for numbers/dates; then require high-overlap match (or allow alias sets for entities).
   · Acceptance criteria: report valid-run coverage; show that correct abstractive answers are no longer disproportionately filtered.
2. **Fully specify CCV as an algorithm (edge cases + ties) and fix docID naming inconsistencies.**
   · Where: Page 3 - Method: 3.3; Page 4 - Eq. (1)–(2).
   · Add explicit fallback when no evidence-valid runs exist; define argmax domain; define tie-breaking for $\texttt{doc}^*$ and for final selection.
   · Acceptance criteria: a reader can reproduce CCV without guessing hidden implementation details.
3. Add answer canonicalization before grouping by “unique answer”.
   · Where: Page 3 - Method: 3.3.
   · Minimal solution: normalized string grouping + alias heuristics.
   · Acceptance criteria: report how often canonicalization merges candidates and whether it changes winners.
4. **Add uncertainty/statistical reliability for the +0.19 SubEM claim.**
   · Where: Page 4 - Experiments: 4.2; Page 5 - Experiments: 4.3.
   · Run paired bootstrap significance test over queries for CCV vs majority vote at $K=20$.
   · Repeat permutation sampling with multiple RNG seeds and report mean±CI for each K.
   · Acceptance criteria: improvement remains positive under repeated sampling; report CI overlaps.
5. **Audit metric sensitivity and disclose answer normalization rules for SubEM/F1.**
   · Where: Page 4 - Experiments: 4.2; Page 4 - Setup.
   · Include a short error analysis showing SubEM-vs-F1 divergence due to surface forms.
6. **Reconcile K-dependent numbers in the quote-constraint ablation.**
   · Where: Page 5 - Experiments: 4.4.
   · Make K explicit for every reported SubEM and ensure consistency with Table 2.

Nice-to-have (high-value improvements)

7. **Broaden evaluation across at least one more dataset, one more generator, and one more retriever.**
   · Goal: show CCV is not NQ/Qwen/Contriever-specific.
8. Test decoding robustness (T=0 vs T>0) and “unstable-query-only” triggering.
   · Goal: clarify when CCV’s compute cost is justified; potentially reduce cost by only applying CCV when answer diversity across permutations exceeds a threshold.
9. **Strengthen related work with a focused taxonomy and explicit contrasts.**
   · Include permutation-robust methods (e.g., Stable-RAG [1]), position-invariant inference modifications (e.g., KV-Fusion [2]), and attribution-faithfulness caveats (e.g., MIRAGE [3]).

ASCII Diagram — Revision Strategy Roadmap

```
[Fix definitions + reproducibility]
  -> edge cases/ties + docID naming + answer normalization
  -> (expected gain) methods are implementable and comparable

[Fix evidence-validity]
  -> remove extractive bias; report coverage + rejection analysis
  -> (expected gain) broader applicability + less evaluation artifact

[Stabilize empirical claims]
  -> bootstrap + multi-seed permutation sampling + metric normalization
  -> (expected gain) confidence that +0.19 is real, not noise

[Extend robustness scope]
  -> new datasets/models/retrievers + T>0 + cost analysis
  -> (expected gain) defensible "robustness" narrative
```

Storyline Options + Writing Outlines

Option A (recommended): "From brittle answer voting to evidence-identity voting"

1. Start with the concrete failure: same top-`N` set, different order → different answers.
2. Explain why majority voting on *answers* is brittle (surface form entropy + correlated wrong answers).
3. Introduce CCV as "vote on permutation-invariant evidence identity (docID)".
4. Preview the two key implementation concerns: citation reliability + evidence-validity filtering.
5. Preview the result: small but consistent gains; biggest impact on unstable queries.

Option B: "Permutation robustness as a measurable reliability metric"

1. Define permutation robustness explicitly (flip rate / accuracy variance).
2. Show that instability is non-trivial on NQ.
3. Introduce CCV as a selector optimized for stability.
4. Present signal diagnostics first (agreement vs correctness), then main accuracy results.

Abstract Outline (complete)

S1 (Problem + setting): State that RAG answers can change when the *order* of a fixed retrieved set is permuted; define the short-context setting (top-`N` passages) and why it matters.

S2 (Gap): State that inference-time permutation ensembling exists, but majority voting on answer strings ignores evidence/citations and is brittle to surface-form variation.

S3 (Method): Define CCV in one sentence: run `K` permutations, require structured citation, map cited indices to permutation-invariant docIDs, and choose the answer whose docID citations are most consistent among evidence-valid runs.

S4 (Key result): Report the main number with full conditions (NaturalQuestions; Qwen3-8B; Contriever top-5; decoding settings) and the delta vs majority voting at a specific K.

S5 (Diagnostic + scope): Report the agreement–correctness diagnostic (with effect size) and explicitly bound applicability (requires structured citations; evidence-validity check; compute cost `K` forward passes).

Introduction Outline (complete)

P1 (Stakes + phenomenon): Define retrieval-permutation sensitivity in short-context RAG and give a concrete example of answer flips. Transition: "This instability makes RAG outputs hard to trust."

P2 (Why it happens): Explain position bias/attention allocation as a mechanism; clearly distinguish long-context "lost-in-the-middle" vs short-context permutation sensitivity. Transition: "Therefore we seek permutation robustness."

P3 (Prior solutions + limits): Summarize (i) training/model-side invariance methods, (ii) inference-time ensembling with majority vote, (iii) attribution/citation faithfulness work; state limitations relevant to this paper. Transition: "These suggest an alternative aggregation signal."

P4 (Core hypothesis): State the tested hypothesis: correct answers tend to show higher docID agreement across permutations; explicitly acknowledge correlated-failure cases (wrong-but-consistent evidence). Transition: "We exploit this as a selector."

P5 (Method teaser): Explain CCV at a high level with one informal definition of score(a); mention evidence-validity filtering and answer normalization. Transition: "We evaluate CCV under a controlled protocol."

P6 (Results preview): Preview the main gain (scoped) + where CCV changes decisions (unstable queries). Transition: "Contributions:"

P7 (Contributions list): Use 2–3 bullets: (1) training-free docID-consistency voting, (2) empirical evaluation + uncertainty reporting, (3) diagnostic calibration of agreement as a quality signal.

ASCII Diagram — Related-Work Taxonomy Tree (Layered)

```
Related Work Taxonomy (Root: Robustness to retrieval/context perturbations)
├── Branch A: Permutation/Order robustness in RAG
│   ├── Leaf A1: Training-/representation-based permutation mitigation [1]
│   └── Leaf A2: Inference-time order invariance modifications [2]
├── Branch B: Inference-time ensembling / self-consistency
│   ├── Leaf B1: Answer-string majority/self-consistency [4]
│   └── Leaf B2: Voting/selection frameworks for RAG hallucination mitigation [5]
└── Branch C: Attribution / citation faithfulness in RAG
    ├── Leaf C1: Model-internal or sentence-level attribution [3][6]
    └── Leaf C2: Citation-failure analysis & mitigation [7]

This paper (CCV) sits between Branch B and C:
- uses inference-time ensembling (B)
- but changes the aggregation statistic to evidence identity via citations (C)
```

Priority Revision Plan

P0 (Must fix before re-submission)

1. **Correctness & definition pass (1 day)**

- Fix docID/locID naming inconsistency and ensure symbol domains are defined (Page 3 - Method: 3.3).
- Add a complete algorithm box specifying edge cases (no valid runs) and all tie-break rules (Page 4 - Eq. (1)–(2)).
- Add answer normalization/canonicalization before grouping.

2. Evidence-validity redesign (1–2 days)

· Replace “answer appears in quote” with a normalization-aware/fuzzy grounding check.
· Report evidence-valid coverage, rejection analysis, and sensitivity to the filter.

3. Reproducibility expansion (0.5–1 day)

· Add implementation details: prompt template, constrained decoding, permutation sampling scheme, determinism, NQ split/version, chunking, normalization rules.

4. Reliability of the headline number (1–2 days)

· Paired bootstrap test for CCV vs majority at `K=20`.
· Repeat permutation sampling with `≥5` RNG seeds for each K; report mean±CI.
· Reconcile any K-dependent numeric ambiguity (quote-constraint section vs K-scaling table).

P1 (High-value experiments / positioning)

5. External validity (2–4 days depending on infra)

· Add at least one more dataset (e.g., a second ODQA benchmark) and one more retriever or generator.
· Add decoding robustness: T=0 vs low T>0.

6. Signal-level diagnostics (1–2 days)

· Add effect size for agreement vs correctness and calibration plots (accuracy vs agreement bucket; AUC).
· Show how discriminability changes with K (supports or refutes the “signal improves with K” story).

7. Related-work tightening (0.5–1 day)

· Reorganize Related Work into a taxonomy aligned with your contribution.
· Add explicit contrasts to permutation-robust (Stable-RAG [1]), order-invariant inference mods (KV-Fusion [2]), and attribution-faithfulness work (MIRAGE [3]).

P2 (Nice-to-have productization)

8. **Compute-aware deployment strategy (1 day)**

· Introduce an “unstable-query-only” trigger (based on answer diversity across a small K) to reduce cost.
· Report wall-clock/throughput for each K.

ASCII Diagram — Experiment Upgrade Plan

```
P0: Fix definitions + evidence-validity
    -> enables fair/robust comparisons
P0: Add bootstrap + multi-seed permutation sampling
    -> validates whether +0.19 is real
P1: Add second dataset/model + T>0
    -> external validity + stress test
P1: Add calibration/effect size for agreement signal
    -> turns diagnostics into actionable insight
P2: Add unstable-query trigger + cost reporting
    -> deployment relevance
```

Experiment Inventory & Research Experiment Plan

Exp ID	Objective/Hypothesis	Setup (data/split/protocol/baselines)	Metrics	Main Outcome	Claim Supported	Current Limitation
E1	Compare CCV vs majority vote under multi-permutation RAG	NaturalQuestions test (3,610 queries); Contriever top-5; Qwen3-8B; greedy decoding T=0; Baselines: vanilla RAG (standard), vanilla RAG (JSON citation), MoA majority vote (K=5,20)	SubEM, F1	CCV 46.37 SubEM vs majority 46.18 at K=20 (+0.19); F1 +0.17	CCV can beat answer-string voting in this setting	Single dataset/model/retriever; no uncertainty for K=20; answer normalization not disclosed; compute cost not reported
E2	K-scaling: does CCV's gain grow with K?	Same as E1; K ∈ {5,10,15,20}	SubEM	Reported deltas: -0.01, +0.10, +0.11, +0.19	"More permutations help" story	No multi-seed permutation sampling; deltas are small; signal-level tests (AUC/effect size vs K) missing
E3	Ablation: strict quote verification vs relaxed constraint	At K=5; compare "with quote" vs "no quote" evidence-validity	SubEM + coverage stats	Coverage increases (96.37%→99.70%) and SubEM improves by +0.64 (as stated)	Evidence-filtering affects effectiveness	K-dependent number clarity needs reconciliation; filter is extractive and may bias toward substring answers
E4	Diagnostic: agreement score correlates with correctness	Same setting as E1	docID agreement score; Mann–Whitney p-value	Mean agreement 3.04 vs 2.85; p-value significant	Agreement carries some quality signal	Effect size/calibration missing; practical predictiveness unclear
E5	Behavioral targeting: CCV differs from majority mainly on unstable queries	Same setting as E1	%unstable queries; %decisions changed	699 unstable queries (19.4%); CCV differs 19.89% of those	CCV acts where majority vote is least reliable	No breakdown of accuracy gains on unstable subset; no cost-aware triggering strategy

Research-Theme Gap Diagnosis

- **New knowledge:** The paper's diagnostic claim ("agreement correlates with correctness") is currently statistically significant but not yet practically quantified (no effect size, calibration curves). The mechanism claim needs boundary conditions (when agreement fails due to correlated misleading evidence).
- **Reproducibility/reusability:** Underspecified algorithm (edge cases/ties), missing answer normalization, and incomplete experimental details make reimplementation risky—especially given small deltas.
- **Impact on practice/understanding:** The method's compute cost is non-trivial (K forward passes). Without a "when to use CCV" strategy (e.g., unstable-query trigger) and cost reporting, practitioners cannot assess value.

Proposed Research Experiments (P0/P1/P2)

P0 Experiments (must)

Target Claim	Hypothesis	Minimal Design	Controls/Baselines	Metrics	Success Criterion	Estimated Cost/Time	Expected Paper-Quality Gain
+0.19 SubEM is real and not sampling noise	CCV remains better than majority under repeated permutation sampling	Repeat permutation sampling with ≥5 RNG seeds for each K; keep retrieval fixed	Majority vote; CCV	SubEM/F1; mean±CI	CCV mean delta >0 and CI excludes 0 (or at least mostly positive across seeds)	Low (hours)	Stabilizes headline claim; reduces reviewer skepticism
SubEM/F1 not formatting artifacts	Normalization makes results more stable and comparable	Publish answer-normalization rules; re-evaluate all methods with same canonicalization	All methods in Table 1/2/3	SubEM/F1 with/without normalization	Delta remains after normalization; narrative matches tables	Low	Addresses metric-sensitivity criticism
Evidence-validity filter is not overly extractive	Normalization-aware/fuzzy grounding increases coverage without harming gains	Replace "answer-in-quote" with normalization-aware match; report coverage + accuracy impact	Current strict filter; relaxed "doc_i only" variant	Coverage; SubEM/F1	Coverage increases and CCV advantage persists	Low–Med	Makes method more general and defensible

P1 Experiments (strongly recommended)

Target Claim	Hypothesis	Minimal Design	Controls/Baselines	Metrics	Success Criterion	Estimated Cost/Time	Expected Paper-Quality Gain
CCV generalizes beyond one model	CCV improves majority vote for another generator	Swap generator (e.g., another open model) keeping retriever fixed	Majority vote; CCV	SubEM/F1; unstable-query gain	Similar trend on second model	Med	Demonstrates external validity
CCV robust under non-greedy decoding	Agreement signal still helps at T>0	Evaluate T=0 vs T=0.2	Majority vote; CCV	SubEM/F1; agreement AUC	CCV not degraded; or provide explanation	Med	Shows method is not greedy-decoding artifact
“Agreement becomes more reliable with K”	Discriminability (AUC) increases with K	For each K, compute AUC of agreement predicting correctness	—	AUC vs K	AUC increases or the claim is revised	Low–Med	Turns interpretation into evidence-backed claim

P2 Experiments (nice-to-have)

Target Claim	Hypothesis	Minimal Design	Controls/Baselines	Metrics	Success Criterion	Estimated Cost/Time	Expected Paper-Quality Gain
Cost-aware deployment	Many queries are stable; CCV can be triggered selectively	Run small K0 (e.g., 5), compute instability, only expand to K=20 if unstable	Always-K majority/CCV	Accuracy vs cost	Similar accuracy at lower average cost	Med	Improves practical relevance

ASCII Diagram — Experiment Upgrade Plan

```
P0: multi-seed permutation sampling + bootstrap
    -> validate small delta
P0: normalization + improved grounding check
    -> reduce artifact risk
P1: new model + T>0
    -> external validity
P1: AUC/effect size vs K
    -> strengthen mechanism story
P2: unstable-query trigger
    -> cost-effective deployment
```

Novelty Verification & Related-Work Matrix

9A. Contribution Novelty Verdict Board

Claim ID	Author Contribution Claim	Key Evidence Papers [n]	Novelty Verdict Tag	Why	Confidence	Required Repositioning
C1	Training-free inference-time method for permutation-robust RAG that aggregates by **document-ID agreement** (citation-consistent voting) rather than answer frequency	[1][2][5][8]	`partially_overlapping`	Permutation/order robustness and inference-time aggregation are established directions ([1][2][5]). Evidence-based reranking ideas also exist ([8]). CCV's residual novelty is using **model-produced citations mapped to permutation-invariant docIDs** as the aggregation statistic, without training/model-internal access.	Medium	Position CCV as a *simple training-free selector* for permutation-ensembled RAG; avoid broad "universal compatibility" framing and emphasize complementarity to model-side invariance.

Claim ID	Author Contribution Claim	Key Evidence Papers [n]	Novelty Verdict Tag	Why	Confidence	Required Repositioning
C2	Evidence-valid filtering using structured citations/quotes; practical relaxation of strict quote constraint	[3][6][7]	`substantially_overlapped`	Attribution/citation faithfulness and citation-failure mitigation are active areas ([3][6][7]) that already discuss self-citation failures and more principled validation. CCV's current substring-based filter is a weaker instantiation and should be presented as an implementation choice, not novelty.	Medium-High	Downgrade novelty language; cite attribution work explicitly; replace extractive "answer-in-quote" test with a normalization-aware grounding check and report coverage/rejection analysis.
C3	Diagnostic claim: citation/docID agreement correlates with correctness and improves with more permutations	[4][8]	`partially_overlapping`	"Consistency as a quality signal" is a known theme in self-consistency/voting ([4]) and evidence aggregation/reranking ([8]). CCV's docID-agreement is a task-specific variant; current evidence is limited (single setting; p-value without effect size).	Medium	Recast as a scoped empirical hypothesis; add effect size + calibration (accuracy vs agreement / AUC) and multi-seed K-scaling to make the diagnostic practically meaningful.

9B. Related-Work Taxonomy Matrix

Taxonomy Layer	Branch/Leaf	Representative Papers [n]	Common Assumptions	Difference vs This Paper	Novelty Risk Signal
Root	Robustness & selection in retrieval-augmented generation	—	Retrieval provides multiple evidence units; generators may be sensitive to perturbations	CCV focuses on *order perturbations* with fixed top-N	—
Branch → Leaf	Permutation/order robustness → Hidden-state / representation aggregation across permutations	[1]	Multiple permutations are generated; model internals accessible	CCV is training-free and uses citation-derived docIDs rather than hidden states	Overlap on problem + "multiple permutations" axis
Branch → Leaf	Permutation/order robustness → Order-invariant inference modifications (cache/architecture-level)	[2]	Decoder-only LMs; inference pipeline can be modified	CCV does not modify KV handling; instead pays K-pass compute	Overlap on "order invariance" objective axis
Branch → Leaf	Inference-time ensembling → Answer-level self-consistency / voting variants	[4]	Multiple samples/runs; voting over answers	CCV votes on *evidence identity* (docID) not answer strings	Overlap on "consistency improves selection" axis
Branch → Leaf	Inference-time ensembling → RAG-specific ensemble voting frameworks	[5]	Multiple agents/runs; voting selects final output	CCV changes the voting statistic to docID agreement and uses evidence-valid filtering	Overlap on "training-free voting for RAG" axis
Branch → Leaf	Evidence-aware selection → Evidence aggregation / reranking in open-domain QA	[8]	Multiple passages; answer candidates can be reranked using evidence signals	CCV adapts evidence-frequency ideas to citation signals and permutation invariance	Overlap on "evidence-based reranking" axis
Branch → Leaf	Attribution & grounding → Model-internal attribution for RAG	[3]	Model internals available; attribution can be computed post hoc	CCV relies on self-citations; may benefit from more faithful attribution	Risk: CCV's citation signal may be unreliable

Taxonomy Layer	Branch/Leaf	Representative Papers [n]	Common Assumptions	Difference vs This Paper	Novelty Risk Signal
Branch → Leaf	Attribution & grounding → Sentence-level in-generation attribution	[6]	Retrieved docs available; attribution is produced during generation	CCV uses coarser doc-level IDs + optional quote checks	Risk: CCV's evidence validation weaker than SoTA attribution
Branch → Leaf	Attribution & grounding → Citation failure analysis & mitigation benchmarks	[7]	Citations can fail even when answers are correct	CCV must bound claims and handle citation failure explicitly	Risk: correlated citation failures can mislead voting

9C. Head-to-Head Comparison Matrix

Ref [n]	Problem/Setting	Method Core	Strongest Overlap Point	Clear Difference	Impact on Final Judgment
[1]	Retrieval-permutation-induced hallucinations in RAG	Multi-permutation generation + hidden-state clustering + cluster-center decoding	Targets same permutation sensitivity phenomenon	Not citation/docID voting; not purely "drop-in" black-box	CCV novelty must be framed as *training-free, citation-based* selector, not "first to handle permutations"
[2]	Position/order invariance in RAG	KV-cache fusion to reduce order dependence	Same goal: reduce order effects	Requires inference pipeline changes; different assumptions	Strengthens need to position CCV as simpler but compute-heavy alternative
[3]	Faithful attribution in RAG; self-citation failures	Uses model internals/saliency for attribution	Directly relevant to citation reliability	Solves attribution, not permutation voting	CCV must acknowledge citation unreliability and improve evidence-validity checks
[4]	Improving self-consistency via ranked voting	Ranked-answer voting rules across runs	Consistency-based selection across multiple runs	Votes on answers, not evidence identity	CCV is incremental: new aggregation statistic (docID) but conceptually close to self-consistency

Ref [n]	Problem/Setting	Method Core	Strongest Overlap Point	Clear Difference	Impact on Final Judgment
[5]	Hallucination mitigation in RAG via ensemble voting	Retrieval voting + response voting (majority)	Training-free multi-run voting	Does not use citations/docIDs	CCV can be positioned as a plug-in replacement for the voting statistic
[6]	Sentence-level in-generation attribution for RAG	Predict reference needs + attribute sentences	Uses citations/attribution signals	Not focused on permutation robustness	Suggests stronger grounding mechanisms than substring quote checks
[7]	Citation failure modes & mitigation	Benchmark + combined citation methods	Highlights citation failure even with good answers	Not a voting method	CCV must handle citation failure and bound "compatibility" claims
[8]	Evidence aggregation for open-domain QA reranking	Strength/coverage-based evidence aggregation to rerank answers	"Vote on evidence" idea exists	Pre-LLM; not permutation-based; no self-citations	Reduces novelty of evidence-frequency logic; CCV's new part is `citation→docID` mapping under permutations

Contribution-level Novelty Conclusion. Based on retrieved literature, CCV's central idea (C1) is best viewed as **partially overlapping**: it combines established multi-run robustness/ensembling themes ([1][2][5]) with an evidence-based reranking intuition ([8]) and contributes a practical, training-free statistic (docID agreement from citations). The evidence-filtering component (C2) is **substantially overlapped** by stronger attribution/citation-faithfulness work ([3][6][7]) and should be repositioned as an implementation detail to improve, not a novelty claim. The diagnostic claim (C3) is **partially overlapping** with prior consistency-as-signal ideas ([4][8]) and needs effect sizes/calibration and broader validation to be decision-relevant.

References

[1] Stable-RAG: Mitigating Retrieval-Permutation-Induced Hallucinations in Retrieval-Augmented Generation 2601.02993

[2] Parallel Key-Value Cache Fusion for Position Invariant RAG 2501.07523

[3] Model Internals-based Answer Attribution for Trustworthy Retrieval-Augmented Generation 2406.13663

[4] Ranked Voting based Self-Consistency of Large Language Models 2505.10772

[5] Mitigating Hallucination on Hallucination in RAG via Ensemble Voting 2603.27253

[6] SAFE: Improving LLM Systems using Sentence-Level In-generation Attribution 2505.12621

[7] Citation Failure: Definition, Analysis and Efficient Mitigation 2510.20303

[8] Evidence Aggregation for Answer Re-Ranking in Open-Domain Question Answering 1711.05116

Scores

Final Score: **5.5/10**.

Post-Revision Target: **[7.0, 7.8]/10**.

Rationale: The core idea is reasonable and practically motivated, but current evidence is narrow and the method definition/evidence-validity design has major reproducibility and validity risks. If the authors (i) fix the filtering/definition issues, (ii) add uncertainty + metric-sensitivity audits for the small delta, and (iii) broaden evaluation across at least one additional dataset/model/decoding setting, the paper could become a solid, useful incremental contribution.

Appendix · Original Paper PDF

The following pages preserve source layout and include review annotations when available.

The appendix below attaches source paper pages and overlays review highlights for ISSUE / SUGGESTION / EVIDENCE objects.

Callout boxes are placed near page margins to avoid covering core content and remain within page boundaries.

End of report · DeepScientist Review Export

CITATION-CONSISTENT VOTING FOR PERMUTATION-ROBUST RETRIEVAL-AUGMENTED GENERATION

FARS
Analemma
fars@analemma.ai

ABSTRACT

Retrieval-augmented generation (RAG) systems are sensitive to the ordering of retrieved documents, with language models exhibiting position bias that causes inconsistent outputs across different document permutations. We propose Citation-Consistent Voting (CCV), a training-free method that improves RAG robustness by aggregating answers based on document-ID agreement rather than answer frequency. CCV prompts the generator to cite supporting documents, maps citations to permutation-invariant document identifiers, and selects answers with the highest citation consistency across multiple permutations. On NaturalQuestions with Qwen3-8B and Contriever retrieval, CCV achieves 46.3[illegible] ing majority voting by +0.19 points at $K = 20$ permuta[illegible] scales monotonically with the number of permutations, confirms that citation agreement correlates significantly ($p = 1.14 \times 10^{-5}$). CCV requires no additional training any RAG system that can produce structured citations.

WARNING: This paper was generated by an automated research system. The code is publicly available.[1]

Index ID: #P01-A01
Full text moved to continuation pages.

1 INTRODUCTION

Retrieval-augmented generation (RAG) has emerg[illegible] as a powerfu[illegible] tual accuracy of large language models by grounding responses in[illegible] 2023). However, RAG systems exhibit a critical vulnerability: sens[illegible] documents. Liu et al. (2023) demonstrated the "lost in the mid[illegible]" phenomenon, where models struggle to utilize information positioned in the middle of the context. More recently, Zhang et al. (2026) showed that even in short contexts with only five documents, order can induce qualitatively different answers, including confide[illegible]

Index ID: #P01-A05
Full text moved to continuation pages.

Existing approaches to address this permutation sensitivity fall in[illegible] methods like Stable-RAG (Zhang et al., 2026) use preference op[illegible] variant to document ordering, but require access to model weights[illegible] methods like Mixture-of-Agents (Wang et al., 2024) generate multiple responses under different document orderings and aggregate via majority voting, but this approach ignores the rich citation structure that modern RAG systems produce.

Index ID: #P01-A02
Full text moved to continuation pages.

We observe that when language models cite evidence for their[illegible] provides a more stable signal than the answer text itself. Our k[illegible] tend to cite the same underlying document across permutations, w[illegible] citation variance. This motivates a new aggregation strategy: rather[illegible] we vote on document-ID agreement among evidence-valid responses.

Index ID: #P01-A03
Full text moved to continuation pages.

We propose Citation-Consistent Voting (CCV), a training-free met[illegible] on document-ID agreement rather than answer frequency. CCV fil[illegible] ity (ensuring the answer is grounded in the cited document), maps[illegible] document IDs, and selects the answer with the highest citation-co[illegible] are:

Index ID: #P01-A04
Full text moved to continuation pages.

[1] https://gitlab.com/fars-a/latent-mode-voting-rag

Annotation Continuation · Source Page 1

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#P01-A01 · SUGGESTION · MAJOR · #R001 · Part 1

Page 1 - Abstract

Problem: The abstract claims a general, system-agnostic improvement ("compatible with any RAG system that can produce structured citations"), but the method as defined relies on (i) the model emitting stable doc indices/citations, and (ii) an "evidence-valid" filter that checks the *answer string appears in the quote* (a strong, format-specific assumption). This makes the applicability narrower than the current wording suggests.

Why it matters: Reviewers will interpret the abstract as promising broad applicability and robustness. If applicability is limited to extractive/substring-friendly answers (or to models that reliably follow a JSON-citation schema), the abstract overstates scope and may be judged as overclaiming.

Actionable fix (tighten scope + add key assumptions): Add one sentence stating the required citation interface and the two evidence-filter variants, and explicitly note that the strict quote constraint is optional and may reduce coverage.

Mentor Revised Version:
"Citation-Consistent Voting (CCV) is a training-free inference-time aggregator for permutation-ensembled RAG that selects answers whose *supporting document identity* is most consistent across retrieved-document permutations. CCV requires the generator to return a structured citation (document index plus an optional quote), maps indices to permutation-invariant document IDs, and optionally filters generations by a lightweight evidence check. On NaturalQuestions with Qwen3-8B and Contriever top-5 retrieval, CCV improves SubEM by 0.19 over answer-string majority voting at $K=20$ permutations, and citation-agreement scores correlate with correctness ($p=1.14times10^{-5}$) under this evaluation setting."

#P01-A02 · SUGGESTION · MAJOR · #R002 · Part 1

Page 1 - Introduction (opening paragraph)

Problem: The introduction motivates permutation sensitivity mainly via "lost in the middle" and a cited Stable-RAG result, but it does not clarify *what exact threat model* you address: (i) position bias under long contexts, (ii) instability under *short* top-k contexts (top-5), or (iii) both. This matters because CCV's mechanism (ensemble over permutations + aggregation) is primarily an inference-scaling strategy; readers will ask when it is worth the extra K passes.

Why it matters: Without a crisp threat model and a measurable definition of "permutation robustness" (e.g., variance of answer accuracy across permutations; probability of answer flip), it is hard to judge whether CCV solves a distinct problem beyond existing permutation ensembling/self-consistency lines of work.

Actionable fix: Add 2–3 sentences at the end of this paragraph defining (a) the setting (top-5 retrieval, short contexts), (b) the robustness metric you target (e.g., accuracy variance / flip rate across permutations), and (c) the practical deployment motivation for K-permutation inference.

Mentor Revised Version:
"In this paper we focus on *retrieval-permutation sensitivity* in short-context RAG: given a fixed top-N retrieved set (we use $N=5$), different orderings of the same set can induce different final answers. We operationalize permutation robustness as the stability of the generated answer and its accuracy across random permutations (e.g., answer flip rate and accuracy variance across K orderings). Our goal is to improve reliability in this setting using inference-time aggregation, without modifying model weights."

#P01-A03 · SUGGESTION · MAJOR · #R003 · Part 1 · CONT.

Page 1 - Introduction (prior work positioning paragraph)

Problem: The paragraph positions "majority vote over permutations" as the main inference-time baseline, but the related-work surface is likely incomplete for attribution/citation-based verification and for other order-invariance approaches. In particular, there are contemporaneous methods that (a) explicitly target permutation sensitivity (e.g., Stable-RAG [1]) and (b) aim for position/order-invariant RAG via inference-time KV-cache fusion [2]. Also, citation/attribution prompting is known to have format errors and hallucinated references; MIRAGE discusses these self-citation failure modes and proposes model-internal attribution [3]. None of these limitations/alternatives are acknowledged in this paragraph, which makes CCV's novelty claim easier to dispute.

Why it matters: Reviewers will ask: "Is CCV just a heuristic variant of self-consistency/ensemble voting? Why should we trust self-citations?" Strong positioning requires explicitly stating what CCV adds beyond (i) majority vote, (ii) hidden-state clustering/decoding, and (iii) attribution

Annotation Continuation · Source Page 1 (cont. 2)

Continuation sheet linked to source highlights. Click marker headers to jump back to the paper region.

#P01-A03 · SUGGESTION · MAJOR · #R003 · Part 2

systems.

Actionable fix: Add a short contrastive sentence per closest family:
· vs Stable-RAG: CCV is training-free and uses *document-ID agreement* as a signal, rather than clustering hidden states and decoding a cluster center.
· vs KV-Fusion: CCV does not change the model architecture or KV-cache handling; it trades compute (K passes) for robustness.
· vs MIRAGE/SAFE-like attribution: CCV uses citations as a *selection statistic*; attribution papers focus on *faithful explanation/grounding*, and they highlight citation-hallucination failure modes you must guard against.

Mentor Revised Version:
"Inference-time robustness can also be pursued via methods that directly model permutation sensitivity (e.g., Stable-RAG [1]) or modify inference to reduce order effects (e.g., KV-cache fusion for position-invariant RAG [2]). Separately, attribution work shows that self-citation prompting can be unreliable and proposes more faithful attribution mechanisms (e.g., MIRAGE [3]). CCV differs in that it is training-free and uses permutation-invariant document identities as an aggregation signal; however, it must also explicitly address citation/attribution failure modes discussed in this literature."

References:
[1] Stable-RAG: Mitigating Retrieval-Permutation-Induced Hallucinations in Retrieval-Augmented Generation 2601.02993

[2] Parallel Key-Value Cache Fusion for Position Invariant RAG 2501.07523

[3] Model Internals-based Answer Attribution for Trustworthy Retrieval-Augmented Generation 2406.13663

#P01-A04 · SUGGESTION · MAJOR · #R004 · Part 1

Page 1 - Introduction (key insight paragraph)

Claim: "Correct answers tend to cite the same underlying document across permutations, while incorrect answers show more citation variance."

Issue: This is a plausible hypothesis, but as written it reads like a general truth. It is also vulnerable to a correlated-failure counterexample: if retrieval contains a popular-but-wrong passage (or the model repeatedly latches onto the same misleading passage), then *wrong answers can also be citation-consistent*. In that case, maximizing doc-ID agreement may amplify systematic errors rather than fix them.

Why it matters: This is the core mechanism justification. If the signal is only weakly discriminative, the method may have limited value or require additional safeguards (e.g., incorporate doc quality/relevance or multi-document support).

Actionable fix: In the introduction, (i) explicitly frame this as an empirical hypothesis tested in Sec. 4.5, (ii) add one sentence acknowledging the correlated-failure case, and (iii) preview a mitigation/diagnostic you already compute (e.g., agreement-score calibration or combining agreement with an evidence-validity/coverage check).

Mentor Revised Version:
"We hypothesize—and test empirically in Sec. 4.5—that, conditioned on having at least one evidence-valid generation, correct answers exhibit higher *document-ID agreement* across permutations than incorrect ones. This signal is not foolproof: if multiple permutations consistently surface the same misleading passage, an incorrect answer may also appear citation-consistent. We therefore use agreement as an aggregation statistic alongside evidence-validity filtering and report diagnostic analyses that quantify when agreement is (and is not) predictive of correctness."

#P01-A05 · SUGGESTION · MAJOR · #R005 · Part 1 · CONT.

Page 1 - Introduction (method summary + contributions paragraph)

Problem 1 (definition gap): The paragraph introduces "evidence validity" and "citation-consistency score" but does not state the exact definition until Sec. 3.2–3.3. For a short paper, readers benefit from a one-sentence formalization here (even informal) so they can understand what CCV optimizes.

Problem 2 (contribution framing): The contribution bullets mix method + a small absolute improvement (+0.19 SubEM) + a correlation test. Given the modest delta, reviewers will scrutinize whether the improvement is statistically reliable and whether the value is mainly diagnostic (the

Annotation Continuation · Source Page 1 (cont. 3)

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#P01-A05 · SUGGESTION · MAJOR · #R005 · Part 2

correlation signal) rather than a new algorithmic capability.

Actionable fix:
· Add one sentence with the score definition in words: "score(a) = number of evidence-valid runs citing the modal docID for answer a".
· Reframe contributions so C2 emphasizes "robustness / stability improvement" and "where it helps" (unstable queries), not only the mean SubEM delta.
· Add an explicit note that the method targets *permutation robustness* rather than improving the underlying single-pass accuracy.

Mentor Revised Version:
"We propose CCV, which aggregates K generations from different document permutations by selecting the answer whose supporting *document identity* is most consistent: for each answer string a, we count how many evidence-valid runs cite the modal underlying document ID for a, and choose the a with the highest count. CCV is designed to improve permutation robustness (stability and accuracy under order perturbations) rather than to outperform the best single-pass prompt. Empirically, CCV yields small but consistent gains over answer-string majority voting on NaturalQuestions and primarily changes decisions on queries with high permutation instability."

- We propose Citation-Consistent Voting, a training-free inference-time method for permutation-robust RAG that exploits citation structure for answer aggregation.
- We demonstrate that CCV improves SubEM by +0.19 over majority voting on NaturalQuestions at $K = 20$ permutations, with gains scaling monotonically with K.
- We validate that document-ID agreement correlates significantly with answer correctness ($p = 1.14 \times 10^{-5}$), confirming that citation consistency is a meaningful quality signal.

2 Related Work

2.1 Position Bias in Language Models

Large language models exhibit systematic biases in how they process information based on its position within the input context. Liu et al. (2023) demonstrated the "lost in the middle" phenomenon, where models perform best when relevant information appears at the beginning or end of the context, with significant performance degradation for information in the middle. This position sensitivity is particularly problematic for retrieval-augmented generation, where the ordering of retrieved documents can substantially affect answer quality. Our work addresses this challenge through inference-time aggregation rather than architectural modifications.

Index ID: #P02-A01
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2.2 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) enhances language model outputs by grounding responses in retrieved documents (Gao et al., 2023). Dense retrieval methods such as DPR (Karpukhin et al., 2020) and Contriever (Izacard et al., 2021) have enabled effective passage retrieval for open-domain question answering. However, RAG systems inherit the position biases of their underlying language models, making them sensitive to the ordering of retrieved documents. Recent work has explored various strategies to improve RAG robustness, including training-based approaches (Zhang et al., 2026) and inference-time methods (Lee et al., 2024).

Index ID: #P02-A02
Full text moved to continuation pages.

2.3 Self-Consistency and Ensemble Methods

Self-consistency (Wang et al., 2022) improves chain-of-thought reasoning by sampling multiple reasoning paths and selecting the most consistent answer through majority voting. This principle has been extended to multi-agent systems, where Mixture-of-Agents (Wang et al., 2024) aggregates outputs from multiple language models to improve response quality. Tang et al. (2023) applied permutation self-consistency to listwise ranking, demonstrating that aggregating predictions across different input orderings can mitigate position bias. Our citation-consistent voting extends this paradigm by using document citations rather than answer frequency as the aggregation signal.

Index ID: #P02-A03
Full text moved to continuation pages.

2.4 Permutation-Robust RAG

Several approaches have been proposed to address permutation sensitivity in RAG systems. Stable-RAG (Zhang et al., 2026) uses preference optimization to train models that produce consistent outputs regardless of document ordering, but requires access to model weights and training data. Tang et al. (2023) proposed permutation self-consistency for ranking tasks, aggregating predictions across multiple orderings. Our approach differs by exploiting the citation structure in model outputs, using document-ID agreement rather than answer frequency for aggregation, and requiring no training.

3 Method

Index ID: #P02-A04
Full text moved to continuation pages.

We propose Citation-Consistent Voting (CCV), a training-free method for improving RAG robustness by aggregating answers based on document-ID agreement rather than answer frequency. Figure 1 illustrates the three-stage pipeline.

Annotation Continuation · Source Page 2

Continuation sheet linked to source highlights. Click marker headers to jump back to the paper region.

#P02-A01 · SUGGESTION · MAJOR · #R006 · Part 1

Page 2 - Related Work (2.1 Position Bias in LMs paragraph)

Problem: The paragraph ends with "Our work addresses this challenge through inference-time aggregation," which risks conflating *position bias in long contexts* with *permutation sensitivity under fixed top-N short contexts*. You cite "lost in the middle", but CCV's experiments are in top-5 contexts; the reader needs an explicit bridge: why is permutation sensitivity the right operationalization of position bias here, and what is the relationship between the two?

Why it matters: Without this bridge, related work feels generic and the motivation chain weakens: the paper claims to fix a phenomenon (position bias) but evaluates another (permutation variance).

Actionable fix: Add one sentence defining the relationship: "Permutation sensitivity is an *instance* of position bias where relative ordering of equally retrieved evidence changes attention allocation; in short contexts this still appears as answer instability." Also consider adding an empirical citation that studies order effects in RAG evaluation (e.g., retrieval robustness/order metric) to justify that order matters beyond long contexts.

Mentor Revised Version:
"While 'lost in the middle' is typically studied in long contexts, a related failure appears in short-context RAG: reordering a fixed top-N retrieved set can shift attention and change the generated answer. We view this *retrieval-permutation sensitivity* as a practical instance of position bias, and we address it via inference-time aggregation across permutations rather than by modifying model architectures."

#P02-A02 · SUGGESTION · MAJOR · #R007 · Part 1

Page 2 - Related Work (2.2 RAG paragraph)

Problem: The paragraph lists retrievers and then says "Recent work … training-based (Zhang et al., 2026) and inference-time (Lee et al., 2024)", but it does not categorize *what* these methods do along the axis relevant to CCV (order-invariance / permutation-robustness). This makes the related work read like a generic RAG overview rather than a targeted positioning.

Why it matters: CCV's novelty/value is about permutation robustness and aggregation signals. The related-work section should quickly teach the reader the existing solution space: (i) make model less order-sensitive, (ii) aggregate multiple runs, (iii) improve citation/attribution faithfulness, (iv) retrieval-side fixes.

Actionable fix: Rewrite 2.2 as a short taxonomy paragraph with 2–3 buckets, and place CCV explicitly as "inference-time multi-permutation aggregation using citation/document-ID consistency". Also, if you keep Lee et al. (2024) as an 'inference scaling' citation, add one sentence explaining how that relates (or does not relate) to permutation robustness.

Mentor Revised Version:
"Prior work on robust RAG spans (i) *model-side* methods that reduce order sensitivity via training or inference modifications, (ii) *inference-time ensembling* that aggregates multiple generations to mitigate instability, and (iii) *attribution/grounding* methods that improve faithfulness of evidence citations. CCV sits in category (ii) but differs from answer-string voting by using permutation-invariant document identities as the aggregation signal, which is complementary to model-side invariance and to attribution-faithfulness improvements."

#P02-A03 · SUGGESTION · MAJOR · #R008 · Part 1 · CONT.

Page 2 - Related Work (2.3 Self-consistency / ensemble paragraph)

Problem: You correctly cite self-consistency/majority voting, but you do not explicitly justify why *answer-string* majority voting is the wrong statistic in RAG, nor do you compare CCV to other non-majority aggregation rules (e.g., ranked voting variants, confidence-weighting, or marginal-based methods). Given your paper's core claim is "vote on evidence not answers," this paragraph should articulate the failure mode: semantically equivalent answers may have different strings; or incorrect answers may be frequent due to shared bias.

Why it matters: This is where the reader should be convinced that CCV is not just "another voting heuristic" but a motivated shift in aggregation signal.

Actionable fix: Add a short diagnostic example (one sentence) and a one-sentence theoretical intuition: "answer strings are not permutation-invariant and have high surface entropy; docIDs are permutation-invariant under mapping." Also cite one recent work that improves self-consistency via richer voting (e.g., ranked voting self-consistency) as a foil to motivate CCV's alternative.

Annotation Continuation · Source Page 2 (cont. 2)

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#P02-A03 · SUGGESTION · MAJOR · #R008 · Part 2

Mentor Revised Version:
"In RAG, majority voting over answer strings is brittle because equivalent answers may appear with different surface forms, while biased retrieval can make an incorrect answer frequent. CCV changes the aggregation statistic from surface form to ***evidence identity***: after mapping citations to permutation-invariant document IDs, agreement becomes comparable across permutations even when answer strings vary. This is complementary to richer self-consistency rules (e.g., ranked-voting self-consistency) that still operate on answers rather than evidence."

#P02-A04 · SUGGESTION · MAJOR · #R009 · Part 1

Page 2 - Related Work (2.4 Permutation-robust RAG paragraph)

Problem: This paragraph is closest to CCV's core topic but still underspecifies the comparison axis. You mention Stable-RAG and permutation self-consistency for ranking, then state "Our approach differs by exploiting citation structure." This is directionally correct, but reviewers will want a sharper "overlap vs residual novelty" statement:
· Stable-RAG explicitly targets retrieval-permutation-induced hallucinations; CCV also uses multiple permutations. So the novelty cannot be "we use permutations" but must be "we use ***document-ID agreement via citations*** as a training-free selector."
· The paragraph does not acknowledge that citation prompting may be unreliable (format errors / citing non-existent sources), which is relevant here and discussed in attribution literature.

Why it matters: Novelty and defensibility hinge on clean differentiation and on acknowledging failure modes of the proposed signal.

Actionable fix: Expand this paragraph by ~3–4 sentences:
(1) one sentence describing Stable-RAG's core mechanism at a high level (hidden-state clustering / cluster-center decoding) and contrasting with CCV;
(2) one sentence describing the reliability risk of self-citations and how CCV mitigates it (evidence-validity filter, or plans for better attribution);
(3) one sentence clarifying that CCV is training-free and thus targets a different deployment constraint.

Mentor Revised Version:
"Stable-RAG [1] addresses permutation-induced hallucinations by aggregating internal representations across permutations (e.g., clustering hidden states and decoding from a dominant cluster center), requiring model-internal access and/or additional machinery beyond simple voting. In contrast, CCV is training-free and uses an externally observable signal—permutation-invariant document IDs obtained from model-produced citations—to select among K generations. Because self-citations can be unreliable, CCV must explicitly validate or calibrate citations (we use an evidence-validity filter and report coverage/diagnostics), and future work can integrate more faithful attribution methods."

References:
[1] Stable-RAG: Mitigating Retrieval-Permutation-Induced Hallucinations in Retrieval-Augmented Generation 2601.02993

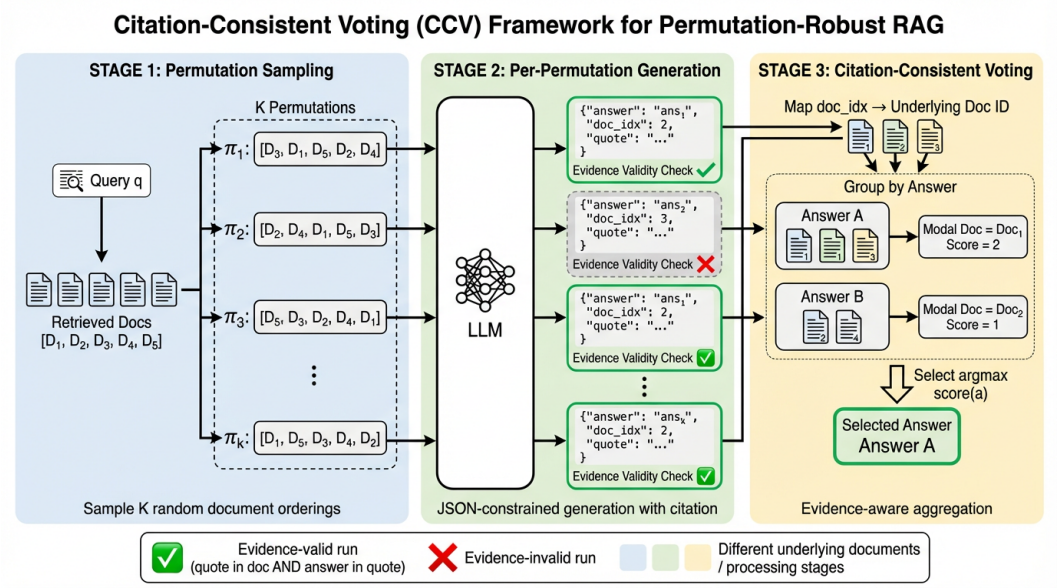


Figure 1: Overview of Citation-Consistent Voting (CCV) for permutation-robust RAG. Given a query and retrieved documents, we generate K responses under different document orderings, filter for evidence-valid outputs (where the answer is grounded in a cited document), and aggregate by document-ID agreement rather than answer frequency.

3.1 Problem Setup

Consider a query q and a set of N retrieved documents $\mathcal{D} = \{d_1, \ldots, d_N\}$. Standard RAG generates a single response by conditioning on a fixed ordering of $\mathcal{D}$. However, the generator's output can vary significantly depending on document order due to position bias (Liu et al., 2023). Multi-permutation RAG addresses this by sampling K random permutations $\pi_1, \ldots, \pi_K$ of the document indices and generating K responses, one for each ordering. The challenge is to aggregate these K responses into a single, more reliable answer.

Index ID: #P03-A01
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3.2 Evidence Filtering

We prompt the generator to produce structured outputs containing an answer, a document index, and a supporting quote. Specifically, each response r_k for permutation π_k contains: (1) an answer string a_k, (2) a document index $j_k \in \{1, \ldots, N\}$ indicating which document in the permuted order supports the answer, and (3) a quote e_k extracted from that document.

A response is considered *evidence-valid* if it satisfies two criteria. First, the cited document index must be valid ($j_k \in \{1, \ldots, N\}$). Second, the answer must be grounded in the cited document, which we verify by checking that the quote e_k appears in document d_{j_k} and that the answer a_k appears in the quote e_k. Responses failing these checks are filtered out before aggregation.

#P03-A02 · ISSUE · MAJOR · #R011 · INDEX · CONT.
Index ID: #P03-A02
Full text moved to continuation pages.

3.3 Citation-Consistent Aggregation

The key insight of CCV is that correct answers tend to cite the same underlying document across permutations, while incorrect answers show more citation variance. We exploit this by aggregating based on document-ID agreement rather than answer frequency.

For each permutation π_k, we map the cited document index j_k to an underlying document ID that is invariant to permutation: $\text{docID}(r_k) = \pi_k(j_k)$. This mapping ensures that the same document is recognized as such regardless of where that document appeared in the permuted order.

#P03-A03 · ISSUE · MAJOR · #R012 · INDEX · CONT.
Index ID: #P03-A03
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Annotation Continuation · Source Page 3

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#P03-A01 · SUGGESTION · MAJOR · #R010 · Part 1

Page 3 - Method (3.1 Problem Setup paragraph)

Problem: The setup introduces multi-permutation RAG and says “the challenge is to aggregate these K responses,” but it does not specify whether K permutations are sampled with replacement, whether the retrieved set is fixed per query, and whether the same retrieved set is used for all permutations (implied) versus re-retrieval. These details matter for reproducibility and for interpreting the robustness claim.

Why it matters: If the retrieved documents differ across runs (query-time nondeterminism, retrieval randomness), then “permutation robustness” becomes conflated with “retrieval robustness”. Your method assumes a fixed set where only order changes.

Actionable fix: Add 2–3 explicit assumptions:
- retrieval is deterministic and the top-N set is fixed per query;
- you sample K random permutations *without replacement* over the same set (or specify if with replacement);
- define a robustness metric (answer flip rate / variance) in this section so the method has a clear optimization target.

Mentor Revised Version:
“For each query, we fix the retrieved top-N document set $mathcal{D}$ (using a deterministic retriever configuration) and generate K answers by sampling K random permutations of the same set (report whether sampling is with/without replacement). Our goal is to improve *permutation robustness*—measured by answer stability and accuracy across these orderings—by selecting a single final answer from the K generations.”

#P03-A02 · ISSUE · MAJOR · #R011 · Part 1

Page 3 - Method (3.2 Evidence Filtering paragraph)

Problem 1 (overly strict / potentially misaligned “evidence-valid” test): The filter requires (a) the quote substring exists in the cited document and (b) the *answer string appears in the quote*. This is effectively an extractive constraint; it will reject many correct but abstractive answers (e.g., normalized entities, multi-token variants, date formats), and it can be gamed by the model copying the answer into any quote-like span.

Problem 2 (terminology/notation inconsistency): You introduce $j_k in {1,ldots,bar N}$ but later validate $j_k in {1,ldots,N}$. The meaning of $bar N$ is not defined, and this creates avoidable confusion about index ranges.

Why it matters: Since CCV’s scores count only “evidence-valid” responses, the filter definition affects both (i) coverage and (ii) which answers win. A too-strict filter can bias results and reduce generality; an easily gamed filter can inflate “validity” without true grounding.

Actionable fixes:
1) Replace “answer appears in quote” with a more robust entailment-style check, or at least a normalization-aware match:
- For entity answers: case-folding + punctuation stripping + alias matching.
- For numeric/date answers: canonicalization rules.
- For multi-token answers: allow partial match with high overlap.

If you want to stay training-free, you can implement lightweight normalization plus a fuzzy match threshold.
2) Define the index domain cleanly and remove $bar N$ unless needed.
3) Report how often this filter rejects otherwise plausible answers and whether rejection correlates with correctness.

Mentor Revised Version:
“A response is evidence-valid if it cites an in-range document index $j_kin{1,ldots,N}$ and provides a quote string e_k that occurs in the cited document. To avoid restricting to purely extractive answers, we check grounding using a lightweight normalization-aware match between the predicted answer and the cited quote (e.g., canonicalizing case/punctuation and normalizing numbers/dates), and we report the resulting valid-run coverage and rejection analysis.”

#P03-A03 · ISSUE · MAJOR · #R012 · Part 1 · CONT.

Page 3 - Method (3.3 Citation-consistent aggregation paragraph)

Problem 1 (notation bug / clarity): You define the mapping as $mathrm{locID}(r_k)=pi_k(j_k)$, but the rest of the section uses $operatorname{docID}(r)$. This looks like a typo/inconsistency (locID vs docID). It becomes more confusing because docID is the *underlying* permutation-invariant identifier.

Annotation Continuation · Source Page 3 (cont. 2)

Continuation sheet linked to source highlights. Click marker headers to jump back to the paper region.

#P03-A03 · ISSUE · MAJOR · #R012 · Part 2

Problem 2 (tie-breaking and answer equivalence): The algorithm groups by “unique answer a” and tie-breaks by raw answer frequency. But in QA, semantically identical answers often differ in surface form (“Barack Obama” vs “Obama”; “United States” vs “USA”), so grouping by raw string can fracture votes and degrade CCV. In the worst case, CCV might select a less canonical alias purely due to string variance.

Why it matters: This is the core algorithm definition. If docID mapping is unclear or answer canonicalization is missing, the method becomes brittle and hard to reproduce.

Actionable fixes:
1) Use one identifier name consistently. For example: define $operatorname{docID}(r_k)=pi_k(j_k)$ and use docID everywhere.
2) Add an answer normalization step before grouping (lowercasing, punctuation stripping, common alias rules), or compute equivalence classes via simple heuristics (e.g., whitespace token set overlap) since you want training-free.
3) Report how often normalization merges answers and whether it changes winners.

Mentor Revised Version:
“For each evidence-valid response r_k, we map its cited index to a permutation-invariant identifier via $operatorname{docID}(r_k)=pi_k(j_k)$. We then normalize answer strings (case/punctuation/alias rules) and group responses by the normalized answer. CCV selects the answer with the largest within-group agreement on the modal $operatorname{docID}$, and uses (normalized) answer frequency only as a secondary tie-break.”

Table 1: Main results on NaturalQuestions (3,610 test queries) with Qwen3-8B and Contriever Top-5 retrieval. Citation-consistent voting achieves the highest SubEM and F1 among multi-permutation methods at $K = 20$.

Method	K	SubEM (%)	F1 (%)
Vanilla RAG (Standard)	1	48.98	41.10
Vanilla RAG (JSON Citation)	1	45.10	45.18
MoA Majority Vote	5	45.82±0.28	46.38±0.21
MoA Majority Vote	20	46.18	46.66
Citation-Consistent Vote (Ours)	20	**46.37**	**46.83**

For each unique answer a, let $R(a)$ denote the set of evidence-valid responses producing answer a. We compute the modal document ID among these responses:

$$\text{doc}^*(a) = \arg\max_d |\{r \in R(a) : \text{docID}(r) = d\}| \quad (1)$$

The citation-consistency score for answer a is then defined as the number of evidence-valid responses that cite this modal document:

$$\text{score}(a) = |\{r \in R(a) : \text{docID}(r) = \text{doc}^*(a)\}| \quad (2)$$

The final answer is selected as $\arg\max_a \text{score}(a)$, with ties broken by [illegible] answer frequency across all K runs.

3.4 Practical Considerations

Quote Constraint Relaxation. The strict quote verification (requiring exact substring match) can reduce citation path coverage when models produce paraphrased or slightly modified quotes. We find that relaxing this constraint to only require valid document index (without quote verification) improves coverage while maintaining the benefit of document-ID agreement. We evaluate both variants in our experiments.

Computational Cost. CCV requires K forward passes through the generator, the same as standard multi-permutation approaches like Mixture-of-Agents (Wang et al. [illegible] of evidence filtering and citation-consistent aggregation is negligible compared to generation cost. The method is compatible with batched inference and can leverage KV-cache sharing across permutations that share document prefixes.

4 Experiments

4.1 Experimental Setup

We evaluate Citation-Consistent Voting on the NaturalQuestions (Kwiatkowski et al., 2019) open-domain QA benchmark, using 3,610 test queries. For retrieval, we use Contriever (Izacard et al., 2021) to retrieve the top-5 passages from Wikipedia. The generator is Qwen3-8B with greedy decoding (temperature=0). We report Substring Exact Match (SubEM), where a prediction is correct if the gold answer appears as a substring, and token-level F1 score.

We compare against two baselines. Vanilla RAG generates a single response without permutation ensembling. MoA Majority Vote (Wang et al., 2024) generates K [illegible] ment orderings and selects the most frequent answer. For multi-permutation methods, we evaluate at $K \in \{5, 10, 15, 20\}$ permutations.

4.2 Main Results

Table 1 presents the main results. Citation-consistent voting achieves 46.37% SubEM and 46.83% F1 at $K = 20$, outperforming MoA majority voting by +0.19 SubEM and +0.17 F1. The improvement is consistent across both metrics, demonstrating that document-ID agreement provides a more

#P04-A01 · ISSUE · MAJOR · #R019 · INDEX · CONT.
Index ID: #P04-A01
Full text moved to continuation pages.

#P04-A02 · ISSUE · MAJOR · #R013 · INDEX · CONT.
Index ID: #P04-A02
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#P04-A04 · ISSUE · MAJOR · #R014 · INDEX · CONT.
Index ID: #P04-A04
Full text moved to continuation pages.

Annotation Continuation · Source Page 4

Continuation sheet linked to source highlights. Click marker headers to jump back to the paper region.

#P04-A01 · ISSUE · MAJOR · #R019 · Part 1

Page 4 - Method (Eq. 1–2 definition block)

Problem 1 (undefined edge cases): Eq. (1)–(2) assume each candidate answer a has at least one evidence-valid response $R(a)neqemptyset$. In practice, you can have queries where *no* run passes evidence filtering (especially under strict quote checks). The paper does not specify the fallback rule, so the algorithm is not fully defined.

Problem 2 (argmax domain + tie handling): In Eq. (1), the domain of d is not stated (is it $din{1,ldots,N}$ doc IDs, or arbitrary identifiers?). Also, ties for $operatorname{doc}^*(a)$ and ties for $operatorname{score}(a)$ are not fully specified (you mention tie-breaking by answer frequency for the final selection, but not for $operatorname{doc}^$).

Why it matters: These are "silent" implementation choices that can materially change results when improvements are small (+0.19). Reviewers will flag missing edge-case definitions as a reproducibility gap.

Actionable fix: Add a short algorithm box or 4–5 lines of text that fully defines:
- docID domain: $operatorname{docID}(r_k)=pi_k(j_k)in{1,ldots,N}$
- what to do if all runs are filtered out (e.g., fall back to majority vote over all K runs, or drop the evidence filter and recompute scores)
- tie-breaking for $operatorname{doc}^*(a)$ and $operatorname{score}(a)$
- (optional but recommended) answer normalization before defining "unique answer a".

Mentor Revised Version:
"We define $operatorname{docID}(r_k)=pi_k(j_k)in{1,ldots,N}$ for each evidence-valid response. For each normalized answer a, let $R(a)$ be the set of valid responses with that answer; if $R(a)=emptyset$ for all a, we fall back to answer-string majority voting over all K runs. Otherwise, we set $operatorname{doc}^*(a)=argmax_{din{1,ldots,N}} |{rin R(a):operatorname{docID}(r)=d}|$ (breaking ties by global frequency), compute $operatorname{score}(a)=|{rin R(a):operatorname{docID}(r)=operatorname{doc}^*(a)}|$, and select the a with maximum score (breaking ties by normalized answer frequency)."

#P04-A02 · ISSUE · MAJOR · #R013 · Part 1

Page 4 - Method/Experiments boundary (3.4 Practical considerations + 4.1 Setup)

Issue: The paper claims CCV is "compatible with any RAG system that can produce structured citations," yet the practical-considerations section already reveals non-trivial engineering assumptions (JSON-constrained output, quote extraction behavior, possible KV-cache sharing across permutations). The experimental setup also fixes decoding to greedy with temperature=0, which reduces output diversity and may understate/overstate the need for voting depending on model.

Why it matters: Reproducibility and external validity. Readers need to know whether CCV's gains persist under:
- non-greedy decoding (temperature > 0),
- different generators/retrievers,
- larger N (top-20),
- contexts where answers are not strict substrings.

Actionable fix: Expand 4.1 to include: exact prompt template (JSON schema), how quotes are extracted, how permutations are sampled, and whether generation is deterministic. Add at least one robustness run varying temperature (e.g., 0 vs 0.2) to show CCV is not an artifact of greedy decoding.

Mentor Revised Version:
"We use a JSON-constrained prompt that requires {answer, doc_idx, quote} outputs; Appendix A lists the exact template. Unless otherwise stated, retrieval is deterministic and we sample K random permutations of the same top-N set per query. We report results under greedy decoding (T=0) and a low-temperature setting (e.g., T=0.2) to assess whether CCV's citation-consistency signal remains predictive when answer strings become more diverse."

#P04-A03 · SUGGESTION · MAJOR · #R020 · Part 1 · CONT.

Page 4 - Experiments (4.1 Experimental setup paragraphs)

Problem: The setup provides high-level choices (dataset size, retriever, generator, greedy decoding, metrics) but is missing several reproducibility-critical details that directly affect small deltas:

Annotation Continuation · Source Page 4 (cont. 2)

Continuation sheet linked to source highlights. Click marker headers to jump back to the paper region.

#P04-A03 · SUGGESTION · MAJOR · #R020 · Part 2

- Exact NaturalQuestions split/version and any preprocessing (answer aliases, punctuation/case normalization).
- Prompt template / JSON schema and any constrained decoding (and how often formatting fails).
- How permutations are sampled (random seed; with/without replacement) and whether retrieval is deterministic.
- Whether the retriever corpus is Wikipedia snapshots consistent with NQ, and how passages are chunked.
- Hardware/latency for $Kin{5,10,15,20}$ so readers can judge the compute–gain trade-off.

Why it matters: With +0.19 SubEM headline improvements, even minor implementation differences (normalization, prompt constraints, random seeds) can change outcomes. Missing details reduce confidence and make it hard for others to adopt CCV.

Actionable fix: Add an "Implementation Details" paragraph (or appendix) that fully specifies the above, and report (i) citation/JSON validity rate and (ii) evidence-valid coverage rate under each filter.

Mentor Revised Version:
"We use the NaturalQuestions open-domain QA test set of 3,610 queries (specify exact split/version) with a fixed Wikipedia snapshot and passage chunking scheme (report chunk size/overlap). For each query we retrieve a deterministic top-N set and sample K random permutations with a fixed RNG seed (state whether sampling is with/without replacement). The generator is prompted with a JSON schema (Appendix A lists the exact template) and decoding constraints; we report schema-valid rate and evidence-valid coverage. All answers are normalized with the same rules before computing SubEM/F1 (list rules). We additionally report wall-clock cost for each K to quantify the compute–robustness trade-off."

#P04-A04 · ISSUE · MAJOR · #R014 · Part 1

Page 4 - Experiments (4.2 Main results paragraph)

Numerical/claim audit (F1/F2): You argue CCV "outperforms MoA majority voting by +0.19 SubEM and +0.17 F1" and then note that Vanilla RAG (standard prompt) has higher SubEM but lower F1 than the JSON-citation prompt, attributing this to "differences in output format rather than method quality." This is plausible, but it is currently asserted without evidence.

Why it matters: The biggest headline is only +0.19 SubEM. If the evaluation metric is sensitive to formatting (and SubEM is substring-based), then small deltas can be artifacts of answer formatting/canonicalization rather than real correctness changes. Also, the claim that the prompt-format difference is not "method quality" needs substantiation.

Actionable fixes:
1) Report canonicalization rules used before computing SubEM/F1 for each prompt type (e.g., lowercasing, punctuation stripping, article removal). If none, add them.
2) Add an error analysis: show examples where JSON citation prompt changes answer surface form causing SubEM drop but F1 rise.
3) Add a statistical test (paired bootstrap) for the +0.19 improvement, or at least multi-seed retrieval/permutation sampling variance, since Table 1 mixes a ± value for K=5 but not for K=20.

Mentor Revised Version:
"To ensure the small gains are not formatting artifacts, we apply identical answer normalization for all methods before computing SubEM/F1 and report the normalization rules. We additionally run a paired bootstrap test over queries for CCV vs majority vote at $K=20$ and provide a brief qualitative analysis of cases where the JSON citation prompt changes answer surface forms (affecting SubEM vs F1 differently)."

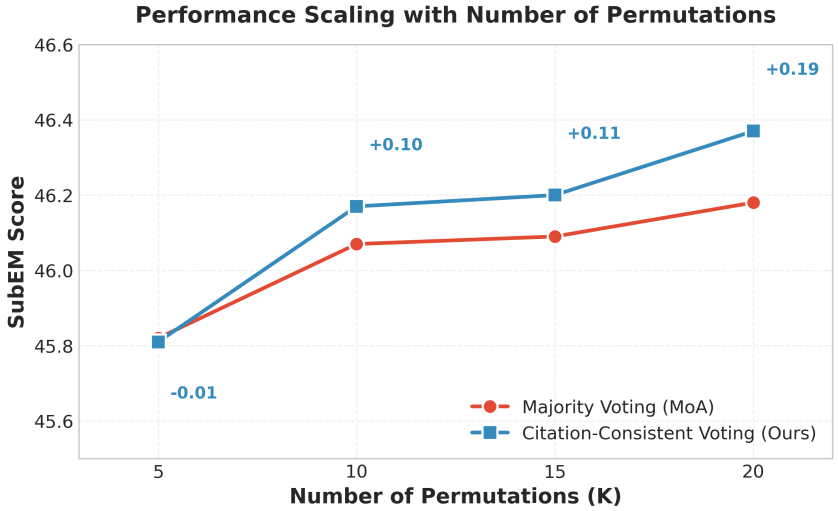


Figure 2: Performance scaling with number of permutations (K) on NaturalQuestions. Citation-consistent voting shows monotonically increasing improvement over majority voting as K increases, reaching +0.19 SubEM at $K = 20$.

Table 2: K-scaling analysis showing SubEM at different numbers of permutations. Δ indicates improvement of citation-consistent voting over majority voting. Improvement scales monotonically with K.

K	MoA SubEM (%)	Citation SubEM (%)	Δ
5	45.82	45.81	−0.01
10	46.07	46.17	+0.10
15	46.09	46.20	+0.11
20	46.18	46.37	**+0.19**

reliable aggregation signal than answer frequency alone. We note that the standard Vanilla RAG prompt achieves higher SubEM (48.98%) but lower F1 (41.10%) compared to the JSON citation prompt, reflecting differences in output format rather than method quality. Among methods using the same JSON citation format, multi-permutation approaches consistently outperform single-pass generation.

4.3 K-Scaling Analysis

Figure 2 and Table 2 show how the improvement of citation-consistent voting over majority voting scales with the number of permutations. At $K = 5$, the two me
($\Delta = -0.01$), but as K increases, citation-consistent voting show
+0.10 at $K = 10$, +0.11 at $K = 15$, and +0.19 at $K = 20$. Th
more permutations provide more reliable citation agreement sig
mechanism benefits from a larger sample of evidence-valid respo
patterns.

#P05-A02 · ISSUE · MAJOR · #R018 · INDEX · CONT.

Index ID: #P05-A02

Full text moved to continuation pages.

#P05-A01 · ISSUE · MAJOR · #R015 · INDEX · CONT.

Index ID: #P05-A01

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4.4 Ablation: Quote Constraint

Table 3 compares citation-consistent voting with and without
$K = 5$. Requiring exact quote matches reduces citation path
of queries with at least one evidence-valid response) and yields
on average. Relaxing this constraint to only require valid document indices increases coverage to 99.70% and valid runs to 4.96, resulting in a substantial improvement from 45.73% to 46.37%

Annotation Continuation · Source Page 5

Continuation sheet linked to source highlights. Click marker headers to jump back to the paper region.

#P05-A01 · ISSUE · MAJOR · #R015 · Part 1

Page 5 - Experiments (4.3 K-scaling analysis paragraph)

Problem: You claim monotonic scaling of the improvement with K and interpret it as evidence that "more permutations provide more reliable citation agreement signals." Two concerns:

(1) Monotonicity is shown only on one dataset/model pair and with small deltas (0.10–0.19). Without confidence intervals across multiple random permutation samples and/or retriever randomness, "monotonic" may be within noise.

(2) The interpretation is one-directional; an alternative explanation is that larger K increases the chance of "finding" a correct answer at least once (a standard inference-scaling effect), and then any selector (including majority vote) improves. To claim the *signal* improves with K, you should show that citation-agreement discriminability (e.g., AUC for correctness vs agreement score) increases with K, not only SubEM.

Why it matters: This paragraph is doing causal interpretation ("agreement becomes more reliable with K"). It should be supported by signal-level diagnostics and uncertainty reporting.

Actionable fix:
· Report variance/CI of SubEM for each K using multiple random seeds for permutation sampling.
· Add a plot/table of agreement-score distributions for correct vs incorrect predictions at each K (or an AUC/KS statistic) to directly support the 'signal reliability' claim.

Mentor Revised Version:
"Across $Kin{5,10,15,20}$ we observe small improvements of CCV over majority voting. To assess whether gains are robust, we repeat permutation sampling with multiple random seeds and report mean±CI for each K. We also measure how well the citation-agreement score separates correct from incorrect predictions at each K (e.g., AUC), which more directly tests whether agreement becomes a better quality signal as K grows."

#P05-A02 · ISSUE · MAJOR · #R018 · Part 1

Page 5 - Experiments (4.4 Quote-constraint ablation paragraph)

F1/F2 consistency issue + interpretation gap: This paragraph states that relaxing the quote constraint improves SubEM from 45.73% to 46.37% "(SubEM +0.64)" at $K=5$. But elsewhere, CCV's main result 46.37% is reported at $K=20$. As written, a reader can easily misinterpret which K each number corresponds to, and whether 46.37 at K=5 is consistent with Table 2 (which reports ~45.81 at K=5 for citation voting).

Why it matters: This is a numerical-clarity/reliability issue. Given that the paper's overall gains are small, any ambiguity about which setting produced 46.37% undermines trust.

Actionable fixes:
1) Make K explicit in the sentence that reports 46.37% (e.g., "at K=5 (no quote) SubEM=XX").
2) Ensure all K-dependent numbers are consistent across Table 1/2/3 and the narrative.
3) If 46.37% at K=5 is a different evaluation variant than Table 2 (e.g., different filtering definition), explicitly label it as such.

Mentor Revised Version:
"At $K=5$, enforcing exact quote matching reduces the fraction of queries with at least one evidence-valid run to 96.37% and yields 4.12 valid runs/query. Relaxing the quote check (keeping only in-range doc indices) increases coverage to 99.70% and 4.96 valid runs/query, improving SubEM by +0.64 *under this $K=5$ setting*. Please ensure that the reported SubEM values for $K=5$ align with Table 2 (or clearly state if this ablation uses a different evaluation variant than the K-scaling table)."

Table 3: Ablation study on quote requirement at $K = 5$. Removing the verbatim quote constraint improves performance by increasing citation path coverage.

Method	SubEM (%)	Citation Path (%)	Avg Valid Runs
MoA Majority Vote	45.71	–	–
Citation Vote (with quote)	45.73	96.37	4.12
Citation Vote (no quote)	**46.37**	99.70	4.96

SubEM (+0.64). This demonstrates that the document-ID agreement signal is more important than strict quote verification for effective aggregation.

4.5 Diagnostic Analysis

To validate the core hypothesis that citation agreement correlates with answer correctness, we analyze the relationship between document-ID agreement scores and prediction accuracy. Correct predictions have significantly higher mean agreement scores than incorrect ones (3.04 vs. 2.85, Mann-Whitney $p = 1.14 \times 10^{-5}$), confirming that citation consistency is a meaningful quality signal.

We further examine where citation-consistent voting differs from majority voting. The method primarily acts on unstable queries, defined as those producing three permutations. Among the 699 unstable queries (19.4% of the test set), citation-consistent voting selects a different answer than majority voting in 19.89% of cases. This targeted behavior confirms that the method intervenes precisely where frequency-based voting is unreliable, leaving stable queries unchanged.

#P06-A01 · ISSUE · MAJOR · #R016 · INDEX · CONT.
Index ID: #P06-A01
Full text moved to continuation pages.

5 Conclusion

We presented Citation-Consistent Voting, a training-free method for improving RAG robustness by aggregating answers based on document-ID agreement rather than answer frequency. On NaturalQuestions, CCV achieves +0.19 SubEM improvement over majority voting at $K = 20$ permutations, with gains scaling monotonically with K. Diagnostic analysis confirms that citation agreement correlates significantly with answer correctness ($p = 1.14 \times 10^{-5}$).

Our work has several limitations. The improvement, while consistent, is modest in absolute terms. Evaluation is limited to a single dataset and model. The method requires K forward passes, multiplying computational cost. Future work could explore combining CCV with training-based approaches like Stable-RAG (Zhang et al., 2026), evaluating on additional datasets, and leveraging other citation signals beyond document-ID agreement.

Index ID: #P06-A02
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Annotation Continuation · Source Page 6

Continuation sheet linked to source highlights. Click marker headers to jump back to the paper region.

#P06-A01 · ISSUE · MAJOR · #R016 · Part 1

Page 6 - Experiments (4.5 Diagnostic analysis paragraph)

Problem: You report a significant difference in mean agreement scores (3.04 vs 2.85) with a Mann–Whitney p-value, and then use this to support "citation consistency is meaningful." Statistical significance is expected with large n (3,610 queries), but the effect size is small. The paper does not report an effect size (e.g., rank-biserial correlation / Cliff's delta) or an operational threshold showing how agreement would be used to *predict* correctness.

Why it matters: The diagnostic claim is one of the three listed contributions. Reviewers will ask: "Is this practically useful, or only statistically detectable?" Without effect size, it is hard to assess practical value.

Actionable fix:
- Report effect size and confidence intervals.
- Add a simple calibration: e.g., correctness rate vs agreement score (bucketed), or an ROC/AUC for agreement predicting correctness.
- Fix the apparent LaTeX/typo in the p-value rendering (it appears corrupted elsewhere in the PDF).

Mentor Revised Version:
"Correct predictions show slightly higher agreement scores than incorrect ones (3.04 vs 2.85; Mann–Whitney $p=1.14times 10^{-5}$). To quantify practical usefulness, we additionally report an effect size (e.g., Cliff's $delta$ with CI) and plot accuracy as a function of agreement score (or AUC for agreement predicting correctness). We also correct the p-value rendering for consistent presentation."

#P06-A02 · SUGGESTION · MAJOR · #R017 · Part 1

Page 6 - Conclusion

Problem: The conclusion repeats the main performance and correlation claims, but it does not clearly restate the scope limits that the paper actually demonstrates: single dataset (NQ), single retriever (Contriever), single generator (Qwen3-8B), and a small absolute delta. It also mentions combining with Stable-RAG as future work but does not articulate a crisp "when to use CCV" recommendation given compute cost.

Why it matters: A defensible conclusion should help the reader form the correct belief: CCV is a lightweight selector that can improve reliability for unstable queries, but it is not a universal accuracy booster and has costs.

Actionable fix: Rewrite the conclusion into (i) validated claim (scoped), (ii) practical guidance (compute vs gain; unstable-query targeting), (iii) prioritized next experiments (multi-dataset, temperature, retriever variance).

Mentor Revised Version:
"In the evaluated setting (NaturalQuestions; Contriever top-5; Qwen3-8B; greedy decoding), CCV provides a small but consistent improvement over answer-string majority voting by selecting answers whose cited document identity is most consistent across retrieval permutations. The method is most relevant for queries that exhibit high permutation instability; in stable cases it typically matches majority voting while still incurring K forward passes. Future work should test CCV across additional datasets/models/retrievers, report statistical reliability under repeated permutation sampling, and explore hybrid designs that combine CCV's citation-based selection with model-side permutation-robust methods such as Stable-RAG."

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